

Dr. Prathmesh Bhadane
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Post-doctoral fellow
Indian Institute of Technology, Bombay, India

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Education			
Degree	Institution	CPI / %	Year
PhD (Materials Engineering)	Indian Institute of Technology, Gandhinagar	8.84/10	2021-2026
MTech (Material Science and Engineering)	Indian Institute of Technology, Gandhinagar	8.00/10	2019-2021
BE (Mechanical Engineering)	Mumbai University	9.18/10	2014-2018

- **Award:** Commendation for **Outstanding Innovation (PhD)** by Indian Institute of Technology, Gandhinagar during the Awards Ceremony of the 15th Convocation.

Post-doctoral Fellow, Chemistry, IIT Bombay (09-Feb-2026 to Present)

- Working on designing **Metal-organic frameworks (MOFs) based heterogeneous catalysts** for gas conversion (CO₂, CH₄) into useful chemicals (Formate, methanol, acetic acid etc.) under the guidance of Prof. Deepti Kalsi (IIT Bombay).
- Teaching assistance for course CH-831L- Instrumentation laboratory, Chemistry at IIT Bombay.

Visiting PhD scholar, University of Birmingham, United Kingdom (04-Sep-2024 to 15 Jan-2025)

- Worked under the joint supervision of Prof. Iseult Lynch and Dr. Swaroop Chakraborty (School of Geography, Earth & Environmental Science, University of Birmingham, UK) on MOF-based environmental remediation and resource recovery.
- Conducted research on Safe and Sustainable by Design MOF Beads for Selective Entrapment of Rare Earth Elements, developing MOF-polymer composite beads for sustainable REE recovery.

PhD research work

- **Research Focus:** Sustainable Nanocomposites and Metal-Organic Framework (MOF) Systems for Environmental Remediation and Resource Recovery.
- **Advisors & International Collaborators:** Worked under the supervision of Prof. Abhijit Mishra (IIT Gandhinagar, India), Prof. Superb Misra (IIT Gandhinagar, India), Prof. Iseult Lynch (University of Birmingham, UK) and Prof. David Fairen-Jimenez (University of Cambridge, UK).

Teaching and Mentoring Experience

- **Assistant Teacher** at Shifa's Group Tutions, Ambarnath, Thane [August 2014-February 2018]
 - Taught Mathematics to students of classes XI and XII, ensuring conceptual clarity and problem-solving proficiency.
- **Private Tutor – Mechanical Engineering Subjects** [July 2016-May 2018]
 - Conducted short courses on core Mechanical Engineering subjects, including Strength of Materials, Heat Transfer, Fluid Mechanics, Machine Design, for a batch of 40 undergraduate students.
- **Independent Instructor**, Materials Characterization Techniques, IIT Gandhinagar [January–May 2024]
- **Research Supervision:** Guided Mr. Adit Rambhia, a final-year B.Tech student at IIT Gandhinagar, on a one-year research project titled "Development of a Reusable Semi-IPN Polymer-Coated Cellulose Acetate Bead Filtration System for Real-World Antibacterial Applications" (2024-2025).

List of Peer-reviewed publications and attended conferences

Peer-reviewed research papers

1. **Prathmesh Bhadane**, D. Menon, P. Goyal, M. R. A. Kiapi, B. K. Satpathy, A. Lanza, I. Mikulska, R. Scatena, S. Michalik, P. Mahato, M. Asgari, X. Chen, S. Chakraborty, A. Mishra, I. Lynch, David Fairen-Jimenez, Superb K. Misra, "A Two-Dimensional Metal-Organic framework for efficient recovery of heavy and light rare earth elements from electronic wastes. Separation and Purification Technology (2024), 130946". **IF: 9.1** DOI: <https://doi.org/10.1016/j.seppur.2024.130946>
2. **Prathmesh Bhadane**, Pankti Dhumal, Emilie Brun, Andrew Britton, Iseult Lynch, and Swaroop Chakraborty. "Safe and Sustainable by Design MOF Beads for Selective Entrapment and Recovery of Rare Earth Elements." Environmental Science & Technology (2025). **IF: 12.2** DOI: <https://doi.org/10.1021/acs.est.5c03112>
3. **Prathmesh Bhadane** and Swaroop Chakraborty, "Cross-Material Synergies of Carbon Nanomaterials, MOFs, and COFs: Innovative Approaches for Sustainable Environmental Remediation and Resource Recovery" Coordination Chemistry Reviews (2025). **IF: 25.6** DOI: <https://doi.org/10.1016/j.ccr.2025.216669>

4.	Dhumal Pankti*, Prathmesh Bhadane *, Bashiru Ibrahim, and Swaroop Chakraborty. "Evaluating the Path to Sustainability: SWOT Analysis of Safe and Sustainable by Design Approaches for Metal-Organic Frameworks." Green Chemistry (2025). (*equal contribution). IF: 10.6 DOI: https://doi.org/10.1039/D5GC00424A
5.	Nature Protocols (2026) : Dhruv Menon, Prathmesh Bhadane , Prateek Goyal, Priya Mahato, Swaroop Chakraborty, Iseult Lynch, Superb Mishra, "An end-to-end workflow for selective heavy metal capture and rare-earth element recovery using metal-organic frameworks". IF:16.0 (Just accepted)
6.	Prathmesh Bhadane , Deepti Kalsi, Rethinking MOFs for REE recovery: Role of functional groups, selectivity limitations and possible future directions", Coordination chemistry Reviews (2026) IF: 25.6 DOI: https://doi.org/10.1016/j.ccr.2026.218197
7.	Prathmesh Bhadane , Deepak G. Prajapati, Adit Rambhia, Mayank Dotiyal, Pramina Kumari Pandey, Deepak Rajput, and Abhijit Mishra. "Antibacterial Activity of Cellulose Acetate-Graphene Oxide Composites: A Comprehensive Multimethod Assessment." ACS Omega (2025). IF:5.2 DOI: https://doi.org/10.1021/acsomega.5c04982
8.	Prathmesh Bhadane , P. Mahato, D. Menon, B. K. Satpathy, L. Wu, S. Chakraborty, P. Goyal, I. Lynch, and Superb K. Misra. "Hydrolytically stable nanosheets of Cu-imidazolate MOF for selective trapping and simultaneous removal of multiple heavy metal ions." Environmental Science: Nano (2024). IF: 5.6 DOI: https://doi.org/10.1039/D3EN00754E
9.	Prathmesh Bhadane , and Swaroop Chakraborty. "Cellulose Acetate-nanoMOF Beads: A Safe, Sustainable and Scalable Solution for Lead Remediation in Complex Water Systems." Environmental Science: Nano (2025). IF: 5.6 DOI: https://doi.org/10.1039/D5EN00056D
10.	Prathmesh Bhadane , and Abhijit Mishra. "The effect of Alkali treatment on Pineapple Leaf Fibers (PALF) on the performance of PALF reinforced rice starch biocomposites." Journal of Natural Fibers (2022): 1-15. IF: 3.1 DOI: https://doi.org/10.1080/15440478.2022.2060406
11.	Prathmesh Bhadane , and Abhijit Mishra. "Versatile, flexible rice starch-graphene oxide bio-nanocomposites." Environmental Science: Water Research & Technology (2024). IF:3.5 DOI: https://doi.org/10.1039/D4EW00419A
12.	Chakraborty, Swaroop, Prathmesh Bhadane , Wilgner Lima da Silva, Hiroto Kitaguchi, Ana Guilherme Buzanich, Sang Pham, Superb K. Misra, and Iuliia Mikulska. "Freeze-drying enables resource-efficient isolation of copper imidazolate metal-organic framework nanosheets for transformation-aware lead capture." Green Chemistry (2026). IF: 10.6 DOI: https://doi.org/10.1039/d6gc03068h
13.	Swaroop Chakraborty, Drexel Roland, Prathmesh Bhadane , Langford Nathan, Dhumal Pankti, Meier Florian, Lynch Iseult, "Integrated Multimethod Approach for Size-Specific Assessment of Potentially Toxic Element Adsorption onto Micro- and Nanoplastics: Implications for Environmental Risk" Nanoscale (2025). IF: 5.1 DOI: https://doi.org/10.1039/D5NR00353A
14.	Chakraborty, Swaroop, Iuliia Mikulska, Rhiannon Boseley, Sang Pham, Prathmesh Bhadane , Pankti Dhumal, Santanu Majumder et al. "Hierarchical Environmental Exposure Transforms Zeolitic Imidazolate Framework-8 and Increases Toxicity in Daphnia magna." ACS nano (2026). IF: 17.3 DOI: https://doi.org/10.1021/acsnano.6c01107
15.	Chakraborty, S., Guilherme Buzanich, A., Prathmesh Bhadane , Kitaguchi, H., Pham, S., & Mikulska, I. (2026). Structural Persistence Masks Commit-Point Chemical Transformation in Copper-Imidazolate Nanosheet Metal-Organic Frameworks. ACS nano (2026). IF: 17.3 DOI: https://doi.org/10.1021/acsnano.6c06011
16.	Saran, Rohitash, Sanchit Saxena, Hritaban Acharya, Prathmesh Bhadane , and Kaling Taki. "Utilization of Water Treatment Plant Sludge for Creating Green Bricks and Examining Its Gamma Radiation Shielding Potential." Journal of Hazardous, Toxic, and Radioactive Waste 29, no. 2 (2025): 04025009. IF: 2.2, DOI: https://doi.org/10.1061/JHTRBP.HZENG-1406
17.	Prathmesh Bhadane , and Mishra Abhijit. "Preparation and characterization of graphene oxide and pineapple leaf fibers reinforced rice starch bio-composite." Materials Today: Proceedings (2022). IF:1.9 DOI: https://doi.org/10.1016/j.matpr.2022.06.092
Selected Conference Presentations	
1.	Prathmesh Bhadane , Abhijit Mishra. "Effect of alkaline treatment of pineapple leaf fibers on the PALF reinforced rice starch composite" International Conference on Natural Fibers 2021, Portugal.
2.	Prathmesh Bhadane , Abhijit Mishra, "Characterization of pineapple leaf fibers and graphene oxide reinforced rice starch bio composite" International Conference on Novel Materials and Technologies for Energy and Environmental Applications, 2022, India.
3.	Prathmesh Bhadane , Abhijit Mishra, "Multifunctional Flexible Starch-Graphene Oxide Bio-Composite" MRSI AGM 2023, IIT BHU and EMRS Spring 2024, Strasbourg, France.

Patent

- **Prathmesh Bhadane**, Goyal Prateek, Mahato Priya and Misra Superb K., “Adsorbent and a process for recovery of rare Earth elements”, Indian Patent Office (IPO), Patent Application No.: 202421047907, Jun. 21, 2024.

Professional Certificates

- Certificate of completion in Professional in Product Design and Analysis from CADD center training services, Mumbai.

Achievements

- IITGN Overseas Research Experience Fellowship – Selected as a Visiting Research Scholar at the **University of Birmingham, United Kingdom** with Funding of 6800 USD.
- Won the Best Poster Award at the University Research Showcase for presenting findings on MOF bead-based water purification system.
- Journal Cover Page Recognition – Featured on the cover page of Environmental Science: Nano and Green chemistry research journals for outstanding research contributions.
- Gold Medalist – Secured first position in the second year of engineering with a 9.83/10 CGPA.
- Qualified Graduate Aptitude Test in Engineering 2021 (GATE 2021) with 98.12%ile.

Position of Responsibility

- Assisted in discipline outreach activities for the Materials Engineering, IIT Gandhinagar, India.
- Teaching Assistant for the undergraduate course "Engineering Graphics"

Technical Skills & Instrumentation Experience

- **Hands-on Experience:** Metal-organic frameworks, Polymer composites, and nanomaterial synthesis, Adsorption and separation studies, Gas conversion experiments, SEM, FTIR, UV-Vis Spectroscopy, DMA, UTM, ICP-OES, TGA and DSC, Contact Angle Goniometer, BET surface area analyzers, XRD and XPS etc.

Reference Persons

- **Prof. Abhijit Mishra (Indian Institute of Technology, Gandhinagar)**- amishra@iitgn.ac.in
- **Prof. Superb Misra (Indian Institute of Technology, Gandhinagar)**- smisra@iitgn.ac.in
- **Prof. Kaling Taki (Indian Institute of Technology, Guwahati)**- kaling.taki@iitg.ac.in
- **Dr. Swaroop Chakraborty (University of Birmingham)**- s.chakraborty@bham.ac.uk